RouteFlow: Final Project Report

Project Team

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Before we begin this project, we would like to sincerely thank Ma'am Zonera for her invaluable guidance and support throughout the *Fundamentals of Software Engineering* course. Her dedication, patience, and passion for teaching have made this learning journey truly meaningful for us. She consistently encouraged us to think critically, work collaboratively, and stay committed to our goals. Her clear explanations and constructive feedback helped us understand the core principles of software engineering with confidence.

We are truly grateful for the time and effort she invested in our growth.

All credit for the successful completion of this project goes to Ma'am Zonera, and we deeply appreciate her mentorship and encouragement.

Thank You, Ma'am Zonera

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Chapter 1

Project Introduction

RouteFlow is a new transport management system for universities that has been created to address the issues that students face while commuting by public transport. Traditional methods of transport management cause delays, miscommunication, and confusion. RouteFlow aims to eradicate these problems by means of smart technology, real-time information, and a central platform to optimize travel to become easier, efficient, and less stressful for students.

Project Overview

In this project, we developed an application that assists students in pre-planning their journeys along with real-time bus route and schedule tracking. Key features include:

- Schedule change Feature
- Online fee management system to check fees and simplify registration.
- feedback system to promote continuous improvement.
- Voting system for seat reservation to manage demand fairly.

Development Process

The project was made in several steps:

Sprint 1

Sprint 1 focused on developing the underlying infrastructure such as a secure login and registration system and a well-designed dashboard to view transportation data.

Sprint 2

Sprint 2 introduced new functionalities such as:

- Fee tracking and challan generation.
- Route updates and access to contact details.
- FAQ feature and feedback submission.

Iteration 3 Plan

During the upcoming sprint, we intend to:

- Refine the user interface.
- Profile management.
- Seat reservation.
- Debug any issues.
- Prepare the application for full deployment.

Chapter 2

User Stories: All Sprints

User stories serve as the foundation for understanding the key functionalities that need to be implemented in RouteFlow. In this sprint, we focused on special features such as fee generation, route updating, and user interaction with their profile. Each user story is broken down into sub-user stories to ensure clarity in requirements and implementation.

2.1 User Authentication (Login/Signup)

User Story: As a student, I want to log in using my credentials so that I can access my bus transportation account.

Sub-User Stories:

- If incorrect credentials are entered, an error message should be displayed to help the user correct their mistake.
- A show/hide password option should be available while typing to ensure accuracy.
- The session should remain active until the user logs out to avoid repeated logins.

User Story: As a new student, I want to sign up by providing my university details so that I can create an account.

Sub-User Stories:

- The system should validate the university ID entered during signup for verification purposes.
- After signing up, the system should send a confirmation email to verify the account.

- The password should meet security requirements to ensure account safety.

2.2 Password Reset Functionality

User Story: As a student, I want to reset my password if I forget it so that I can regain access to my account.

Sub-User Stories:

- The system should send a password reset link to the registered email to allow secure password recovery.
- The new password should meet security standards to ensure protection.
- A confirmation message should be displayed after a successful password reset.

2.3 Dashboard and Navigation

User Story: As a student, I want to access the home page after logging in so that I can navigate through the bus reservation system.

Sub-User Stories:

- The home page should display a personalized welcome message with the user's name.
- Quick access to bus schedules and routes should be available on the dashboard.
- The system should provide a clear and intuitive layout for easy navigation.

User Story: As a student, I want a dashboard that provides an overview of my transportation details so that I can quickly access important information.

Sub-User Stories:

- The dashboard should display the registered bus route and timings.
- A sidebar menu should be available for smooth navigation between different sections.
- A dark mode option should be provided for a customizable user experience.

2.4 Secure Logout Functionality

User Story: As a student, I want a logout button on the dashboard so that I can securely log out of my transportation account.

Sub-User Stories:

- A logout confirmation prompt should appear to prevent accidental logouts.
- After logout, the system should redirect the user to the login page.
- The session should be terminated completely after logging out to ensure account security.

2.5 Contact

User Story: As a Student, I want to view the contact information of the bus service so that I can reach out for inquiries or assistance.

2.6 Feedback

User Story: As a user, I want to rate the bus service from 1 to 10 so that I can share my experience and satisfaction level.

Sub-User Stories:

- A scale from 1 to 10 is displayed.
- User can select only one rating.
- A confirmation appears after submitting the rating.

2.7 Reserve Seat

User Story: As a user, I want to view and select available seats, and receive confirmation after reserving one, so that I can successfully book a seat on the bus.

Sub-User Stories:

- Reserved seats are marked and unavailable.
- Users can select only one available seat.
- A visual seat layout shows seat positions.
- A confirmation message appears after booking

2.8 FAQ

User Story: As a student, I want to access a Frequently Asked Questions (FAQ) page so that I can quickly find answers to common queries.

Sub-User Stories:

- Dropdown list of frequently asked questions is available.
- Dropdown expands and collapse on clicks.
- FAQ page is easily accessible and responsive.

2.9 Fee Details

User Story: As a student, I want to view my fee details so that I can check my payment status.

Sub-User Stories:

- Table of past and current fees is displayed.
- Important information of the fees such as id, semester, due date and amount shown.
- Checkboxes indicating if fees is paid or not.

2.10 Generate Fee

User Story: As a student, I want to generate a fee challan so that I can pay my outstanding dues.

Sub-User Stories:

- Pending fees are displayed before generating challan.
- Option for updating route is available before generating challan.
- The challan can be downloaded in PDF format.

2.11 Profile Page - View Profile Information

User Story: As a student, I want to view my profile information so that I can verify my details.

Sub-User Stories:

- Display full name, contact, gender, route, and email fields in a structured form.
- Allow the user to click "Edit Profile" to enable editing of their information.
- Load and display the existing profile information automatically when the page opens.

2.12 Route Status Page - Live Bus Updates and Details

User Story: As a user, I want to view live status of buses, along with driver and bus details, and current weather conditions so that I can stay informed and safe.

Sub-User Stories:

- Display live location and status (on-time, delayed) of each bus in real-time.
- Show driver name, contact number, and bus details (e.g., bus number, route) alongside live status.
- Display current weather conditions for the bus route or stop location.

2.13 Routes Page - View and Filter Bus Routes

User Story: As a user, I want to view all bus routes with start and end times, and filter them by time or location so that I can plan my travel efficiently.

Sub-User Stories:

- Display a complete list of all bus routes with their respective start and end timings.
- Provide a filter option to view routes based on selected departure or arrival times.

2.14 Schedules Page - Filter Schedules by Route

User Story: As a student, I want to filter schedules by route so that I can quickly find relevant buses.

Sub-User Stories:

- As a student, I want to filter bus schedules by route name so that I can see all buses operating on a specific route.

- As a student, I want the filtered bus schedules to display route details (i.e Availability) so that I can make informed decisions about my travel plans.
- As a student, I want the filtered bus schedules to be sorted by departure time so that I can find the next available bus easily.

Chapter 3

Design

In Sprint 2, the design phase focuses on key diagrams to outline system interactions and structure. The **use case diagram** defines user actions, while the **sequence diagram** illustrates the flow of these actions within the system. The **class diagram** structures the core components, ensuring efficient data handling and interaction. Together, these diagrams provide a clear blueprint for developing and integrating new features.

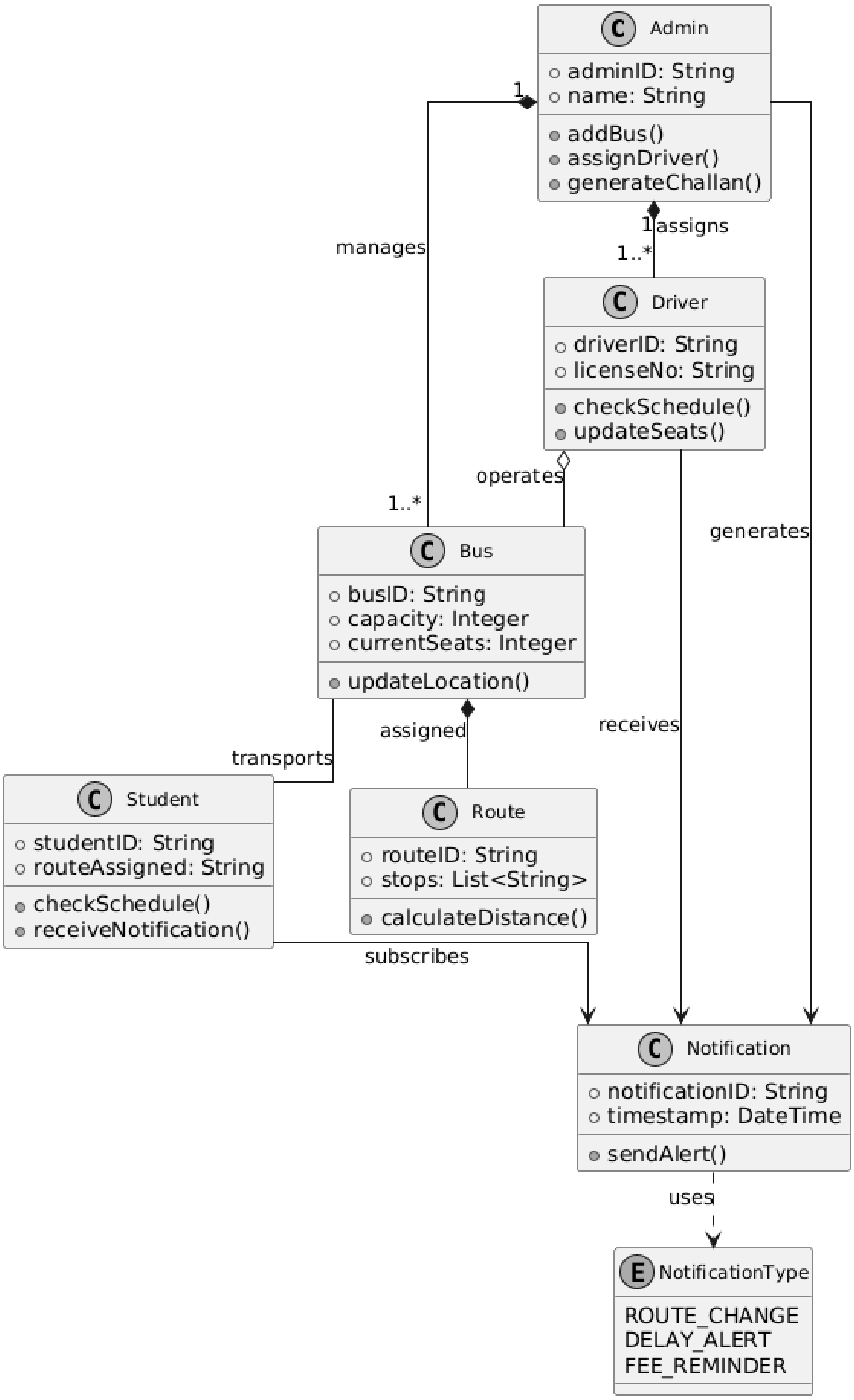

Figure 3.1: Class diagram showing an overview of Project

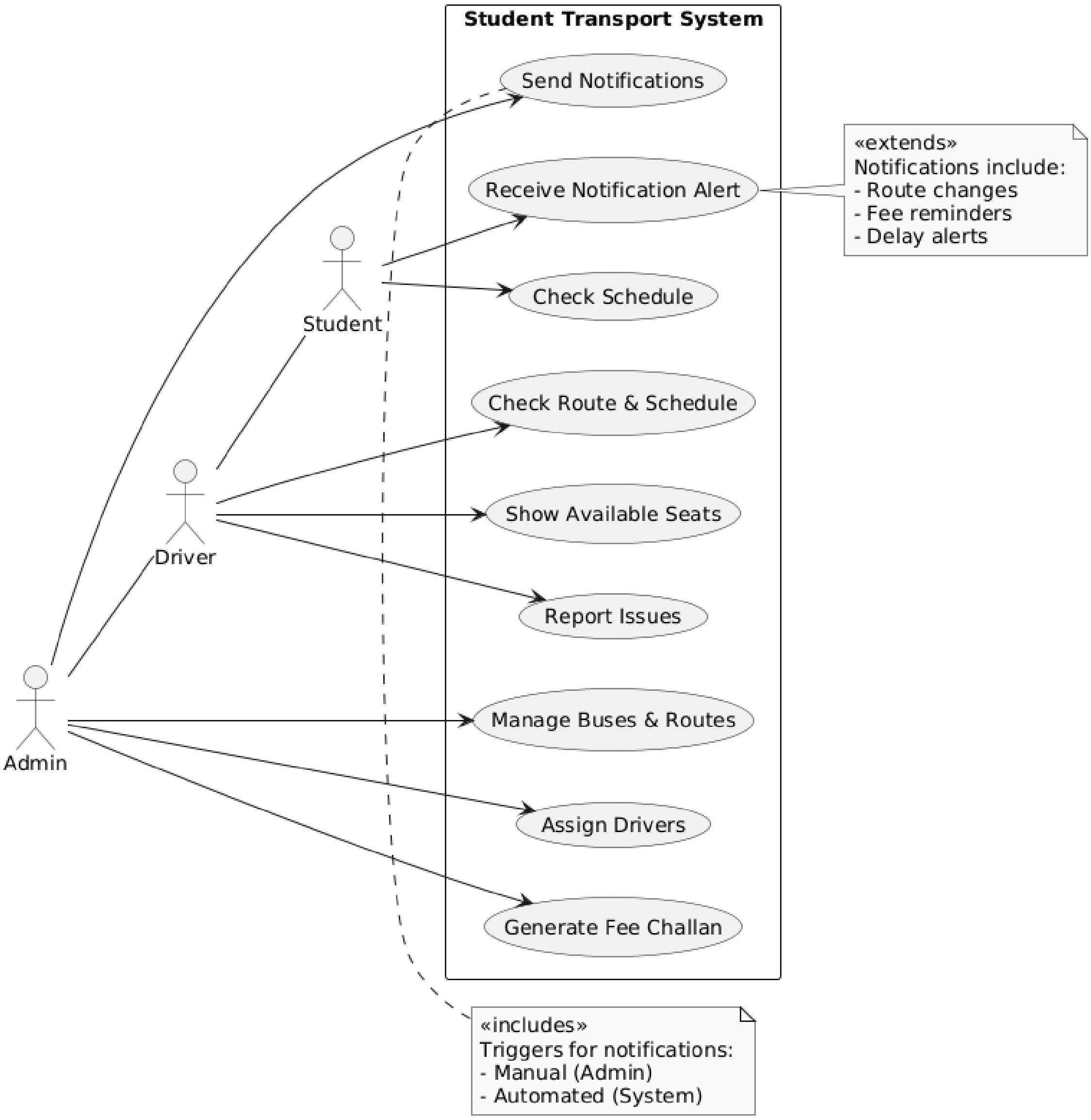

Figure 3.2: Use Case Diagram of Project

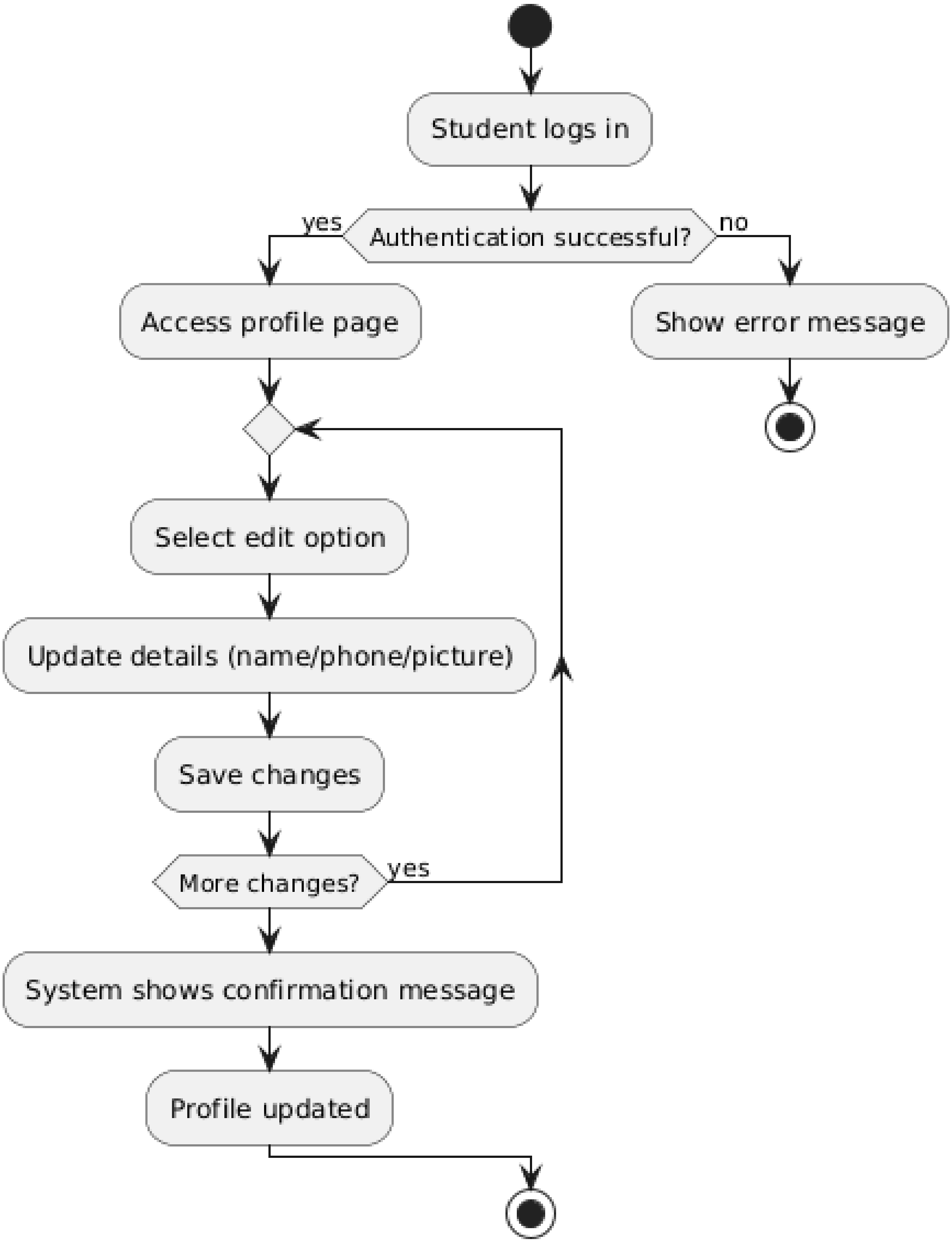

Figure 3.3: Activity Diagram 1: View and Edit Profile

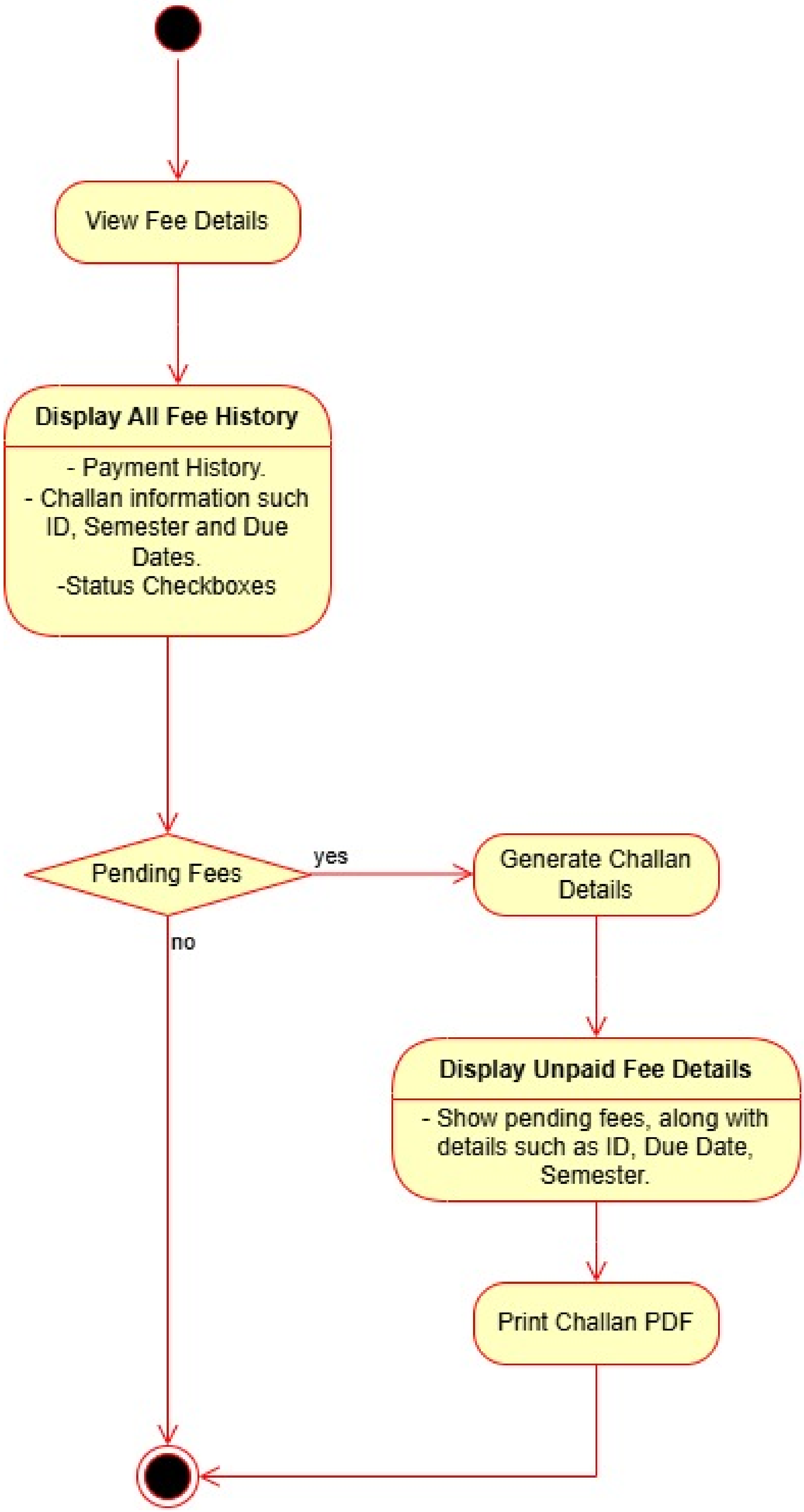

Figure 3.4: Activity Diagram 2: View Fee Details

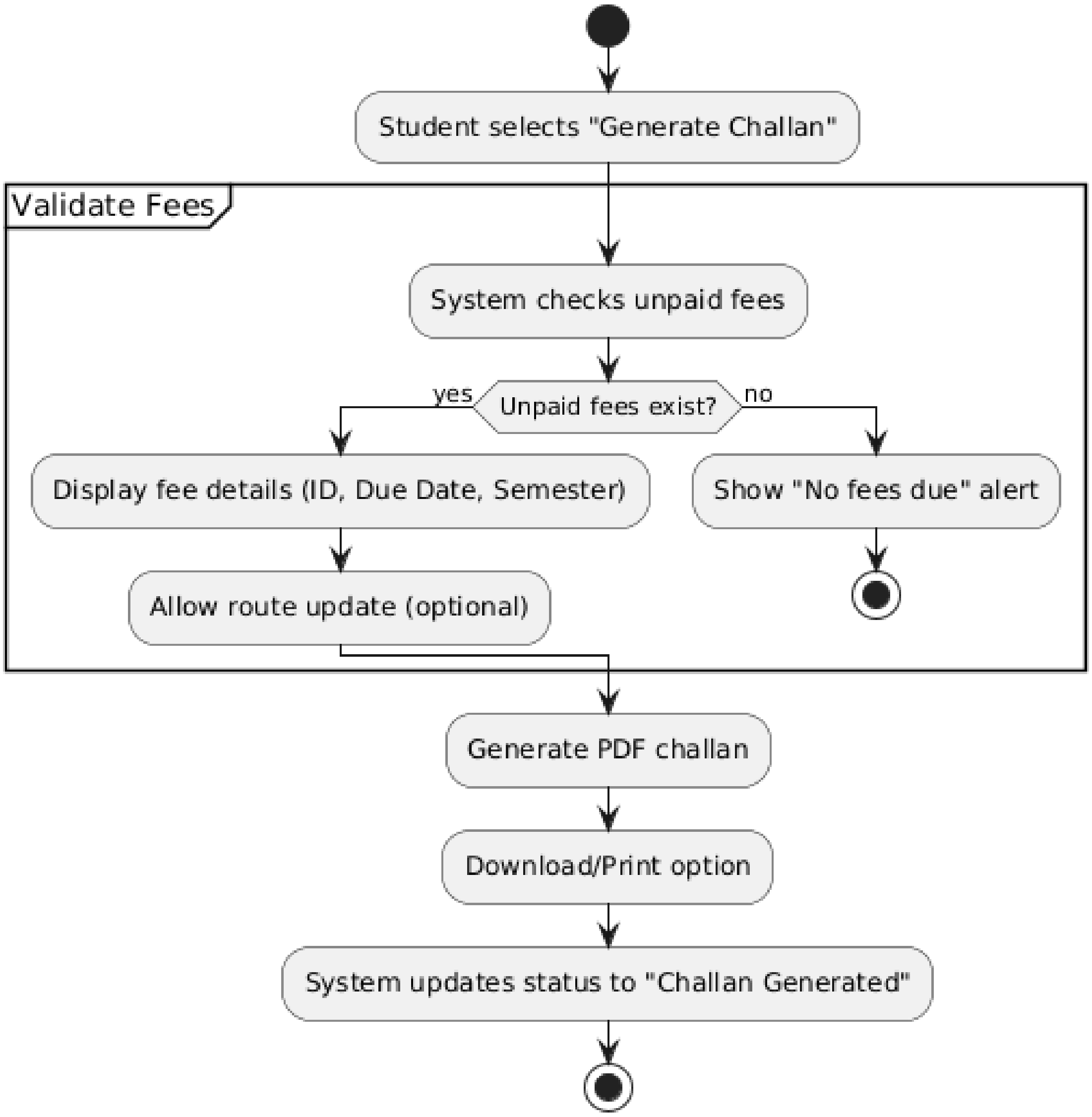

Figure 3.5: Activity Diagram 3: Generate Fee Pdf

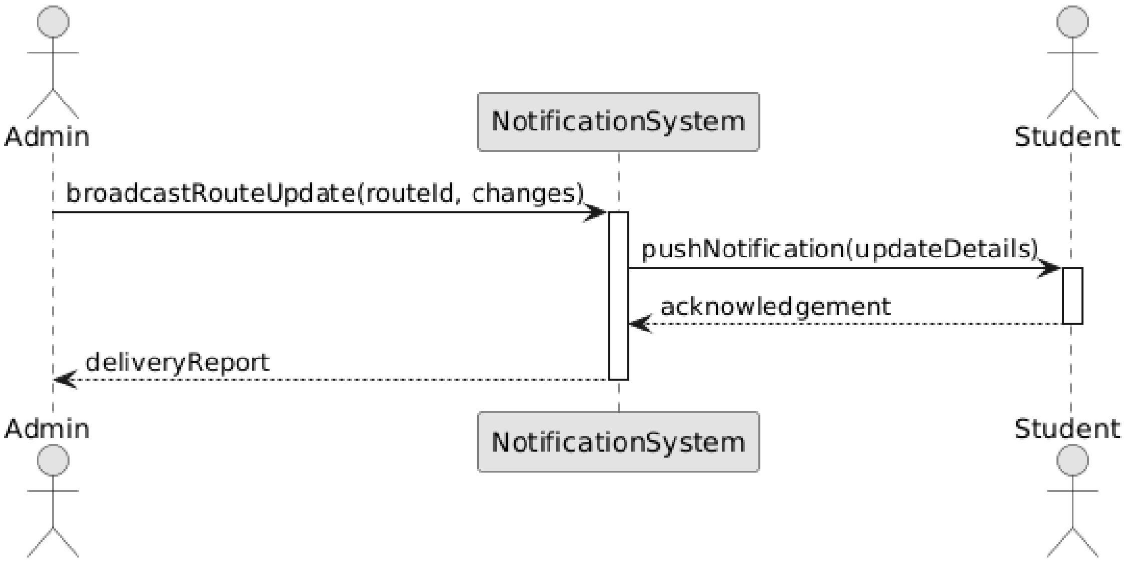

Figure 3.6: Sequence Diagram 1: Notification System

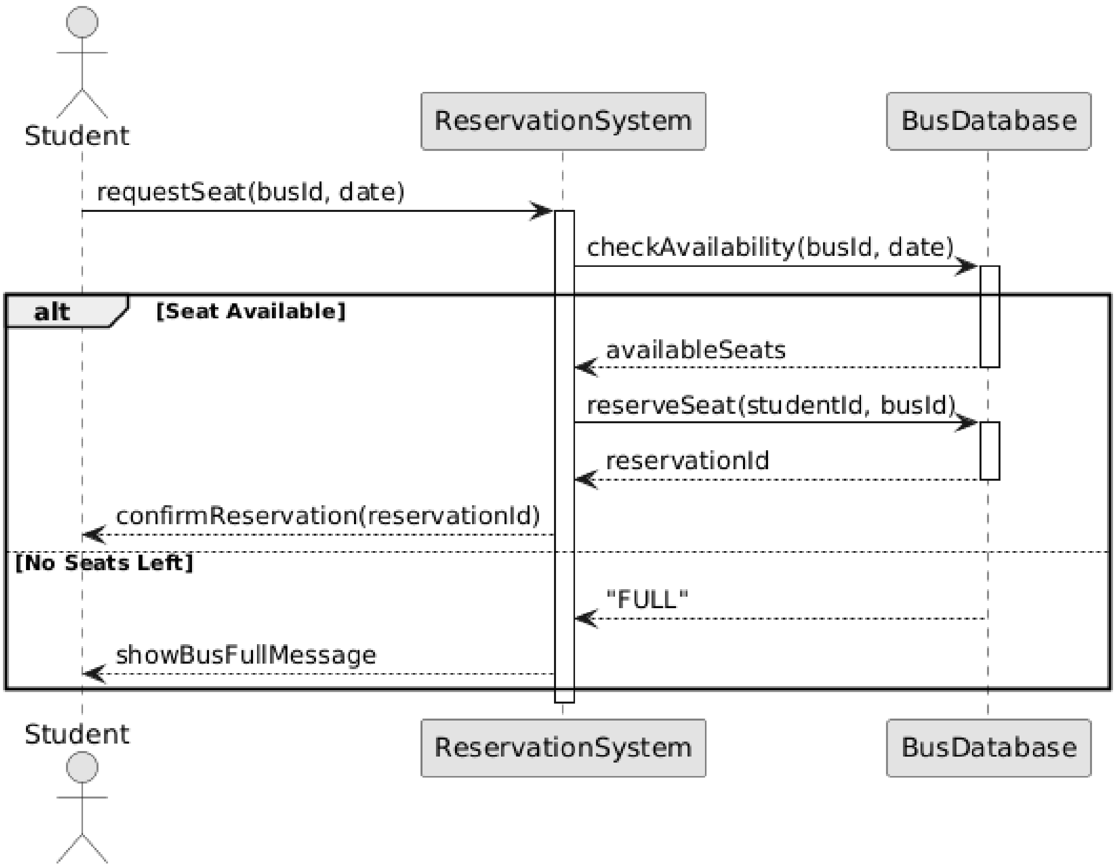

Figure 3.7: Sequence Diagram 2: Reservation System

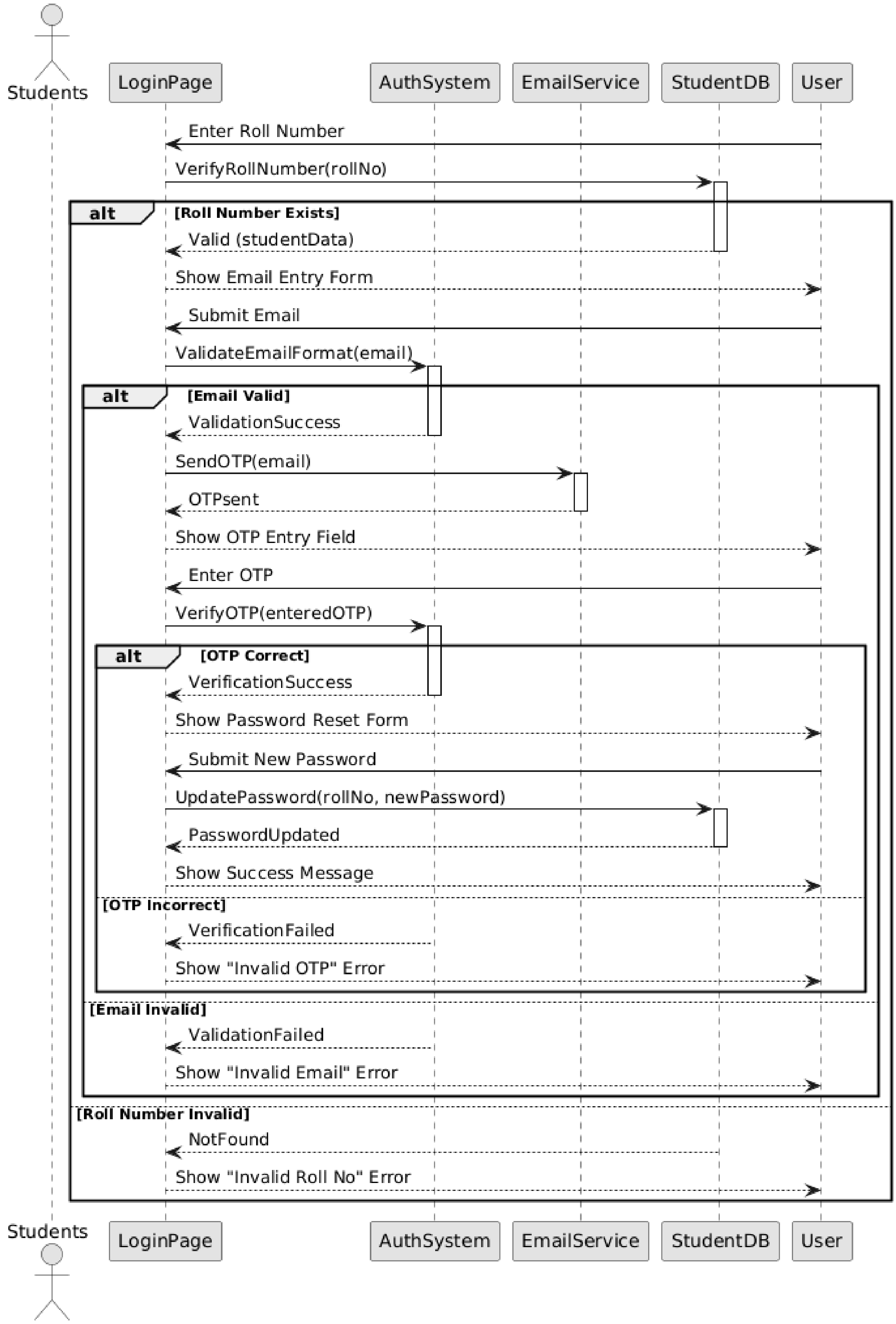

Figure 3.8: Sequence Diagram 3: Authentication System

Chapter 4

Bus Management System Architecture

4.1 Why MVC is Ideal

- **Separation of Concerns:** Clearly separates data handling, business logic, and UI components.
- **Maintainability:** Changes in one layer don't affect other layers (e.g., updating UI doesn't require changing business logic).
- **Testability:** Each component can be tested independently.
- **Flexibility:** Windows Forms interface with controllers, which then communicate with models.
- **Scalability:** Easy to add new features or modules as the system grows.

4.2 Key Components

4.2.1 Presentation Layer (Views)

Consists of Windows Forms such as:

- Home
- Authentication
- Schedules
- Reservations
- Fee Details

– Feedback

These forms collect user inputs and display necessary data.

4.2.2 Business Logic Layer (Controllers)

Controllers process user actions from the Views and interact with Models to perform CRUD operations:

– **Authentication Controller:** Handles user login and registration.

– **Route Controller:** Manages bus routes and schedules.

– **Reservation Controller:** Processes seat bookings.

– **Fee Controller:** Handles fee calculations and challan generation.

– **User Controller:** Manages student and admin profiles.

– **Feedback Controller:** Processes user feedback and suggestions.

4.2.3 Data Access Layer (Models)

Models represent data structures and provide methods to access and manipulate the database:

– **User Model:** Manages student/admin accounts.

– **Route Model:** Stores bus route information.

– **Bus Model:** Contains bus details and availability.

– **Reservation Model:** Stores seat booking records.

– **Fee Model:** Manages fee structure and payment records.

– **Feedback Model:** Stores user comments and ratings.

4.2.4 Services (Additional Functionalities)

Services provide specific background functionalities to the system:

– **Notification Service:** Sends alerts for booking confirmations, fee reminders, and other notifications.

– **Report Service:** Generates reports such as fee challans, booking summaries, and user activity reports.

4.3 Complete Overview Diagram

This diagram shows the layered structure of the Bus Management System using the MVC (Model-View-Controller) architecture. It separates the system into Presentation (Views), Business Logic (Controllers), Data Access (Models), and Services, all connected to a central SQL database for data storage.

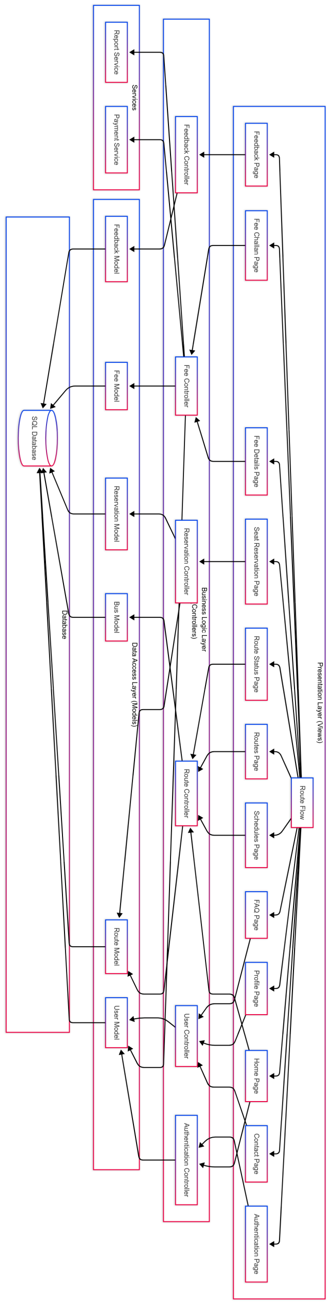

Figure 4.1: MVC Architectural Model

Chapter 5

Actual Implementation Screenshots

Figure 5.1: Landing Page

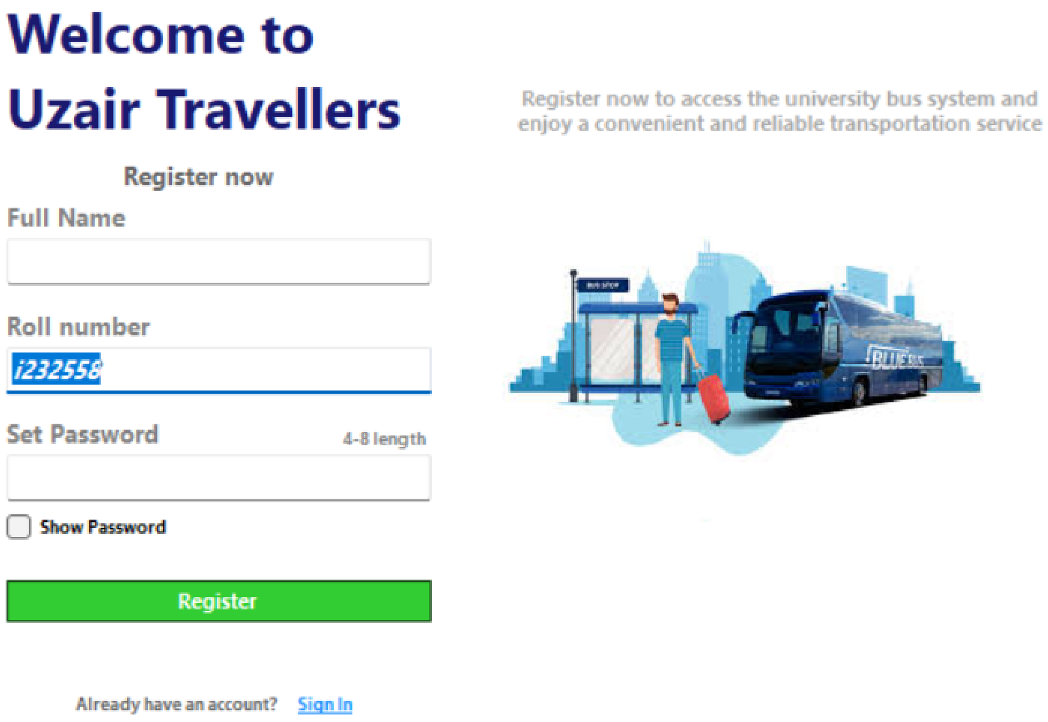

Figure 5.2: Sign Up Page

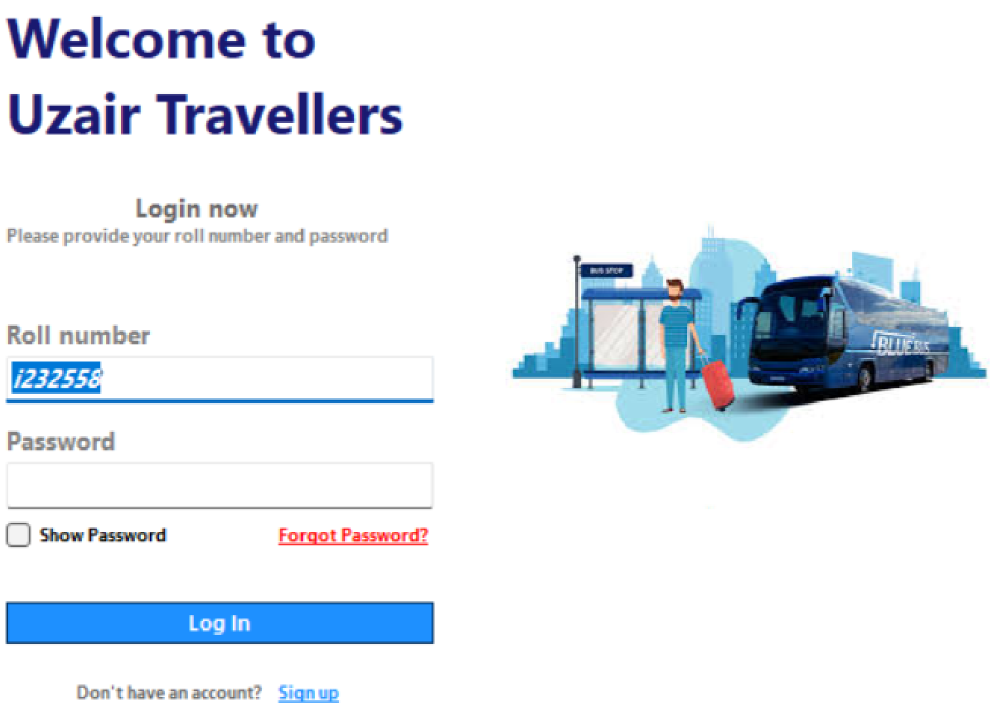

Figure 5.3: Login Page

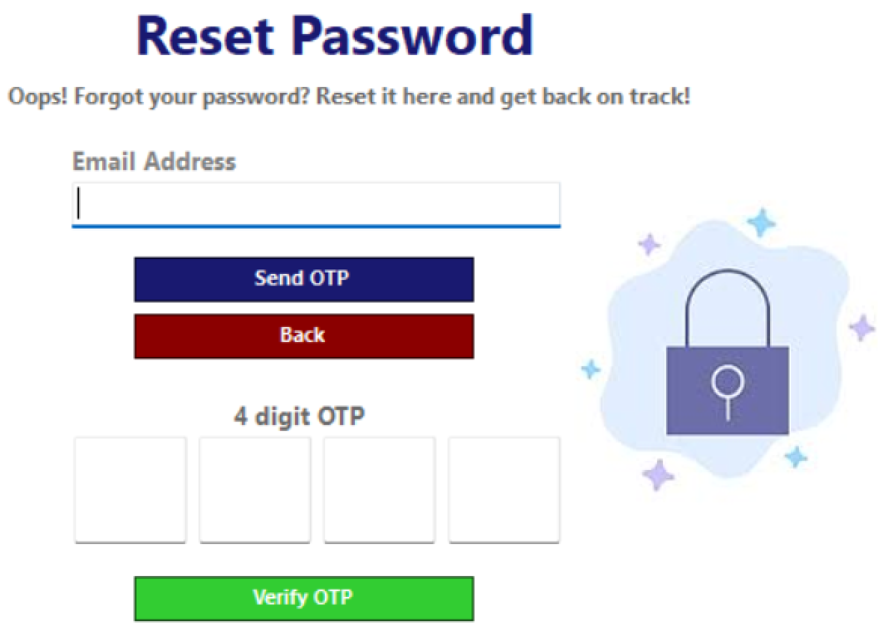

Figure 5.4: Reset Password Page

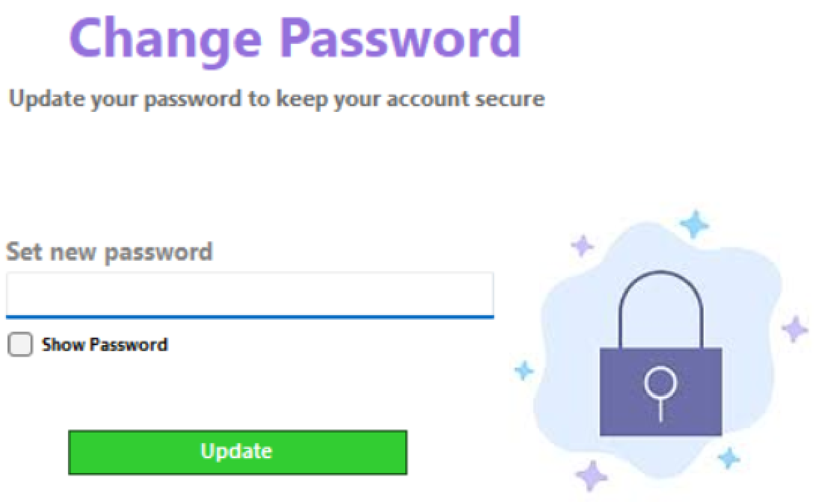

Figure 5.5: Change Password Page

Figure 5.6: Dashboard Page

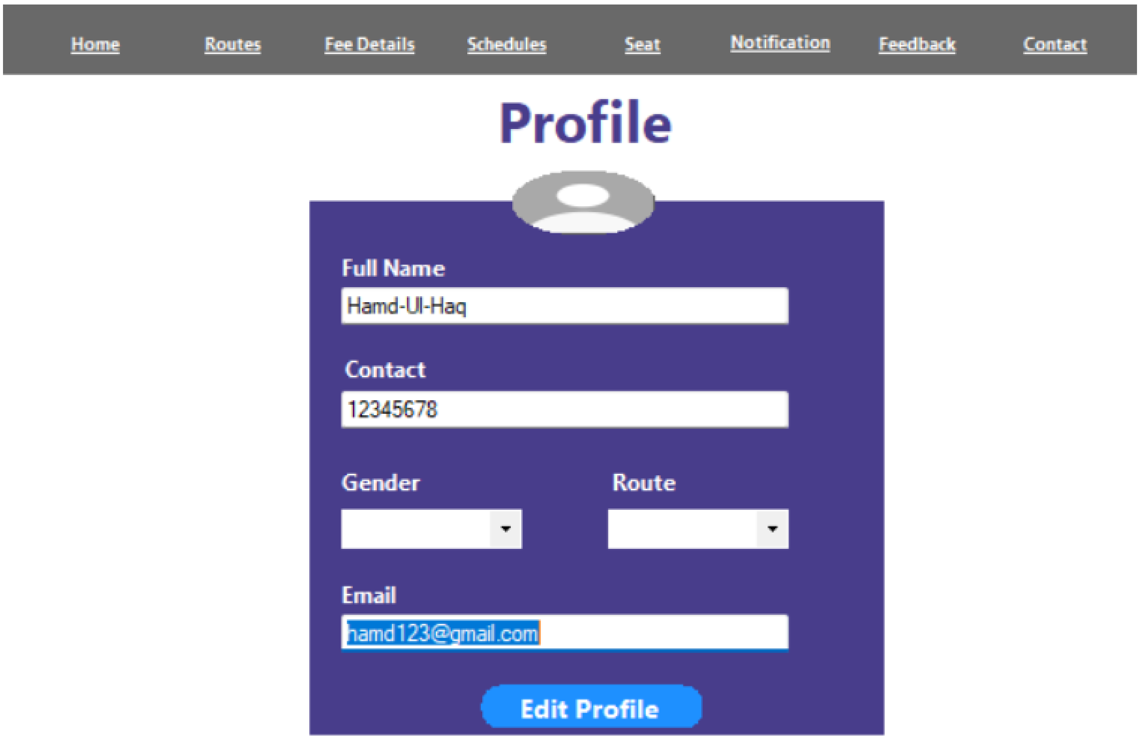

Figure 5.7: Profile Page

Figure 5.8: Route selection Page

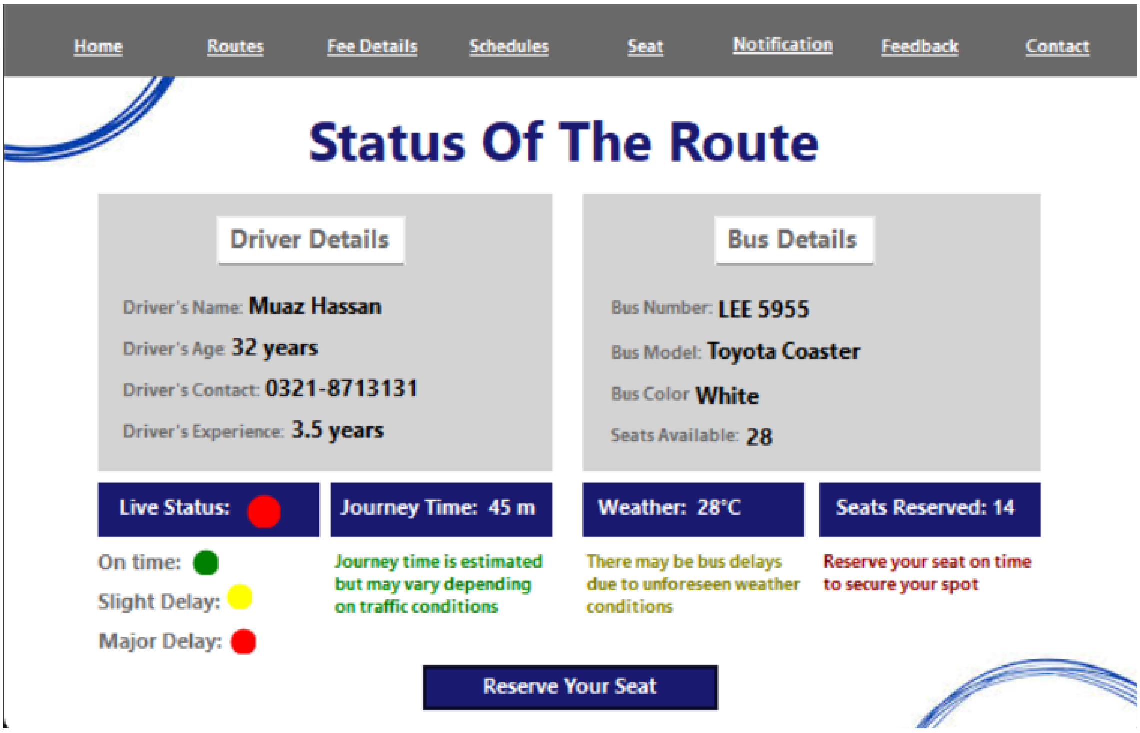

Figure 5.9: Status Of Route Page

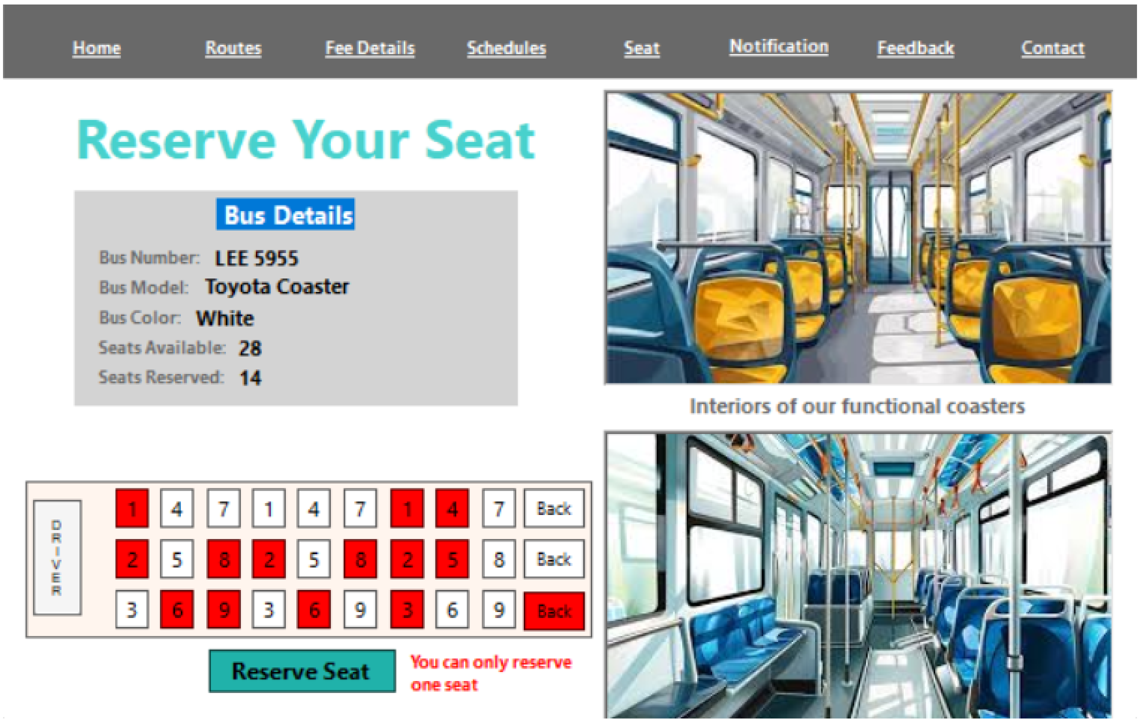

Figure 5.10: Seat Reservation Page

Figure 5.11: Fee Details Page

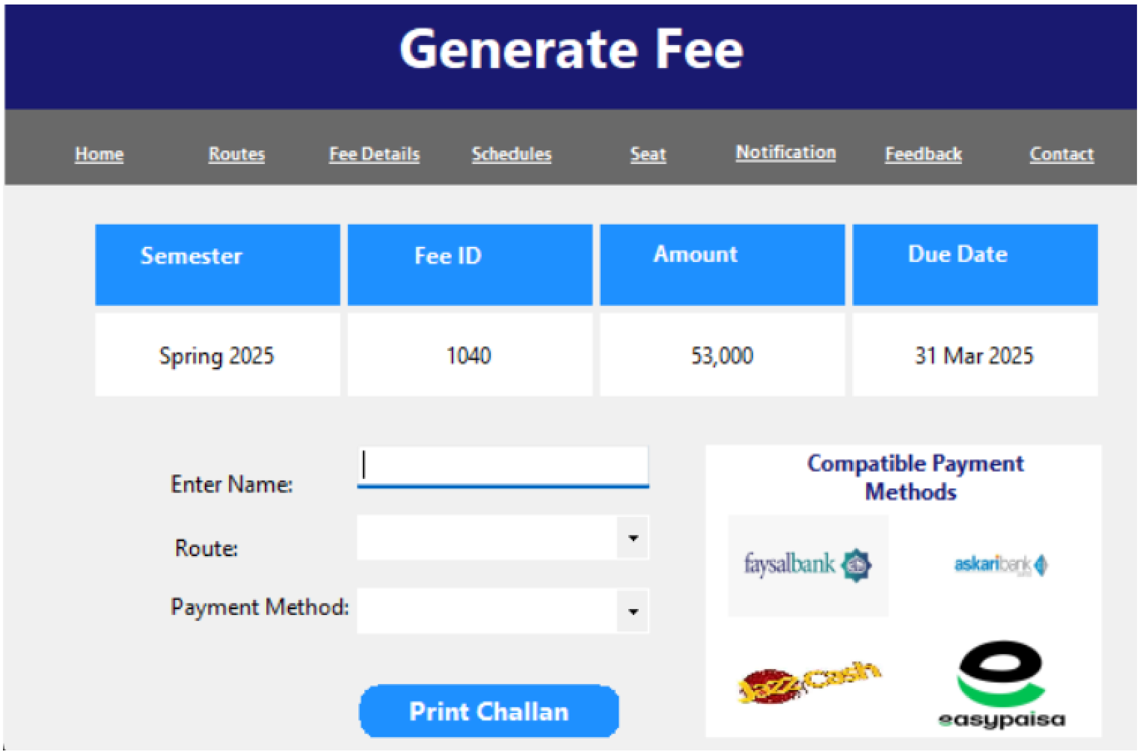

Figure 5.12: Generate Fee Challan Page

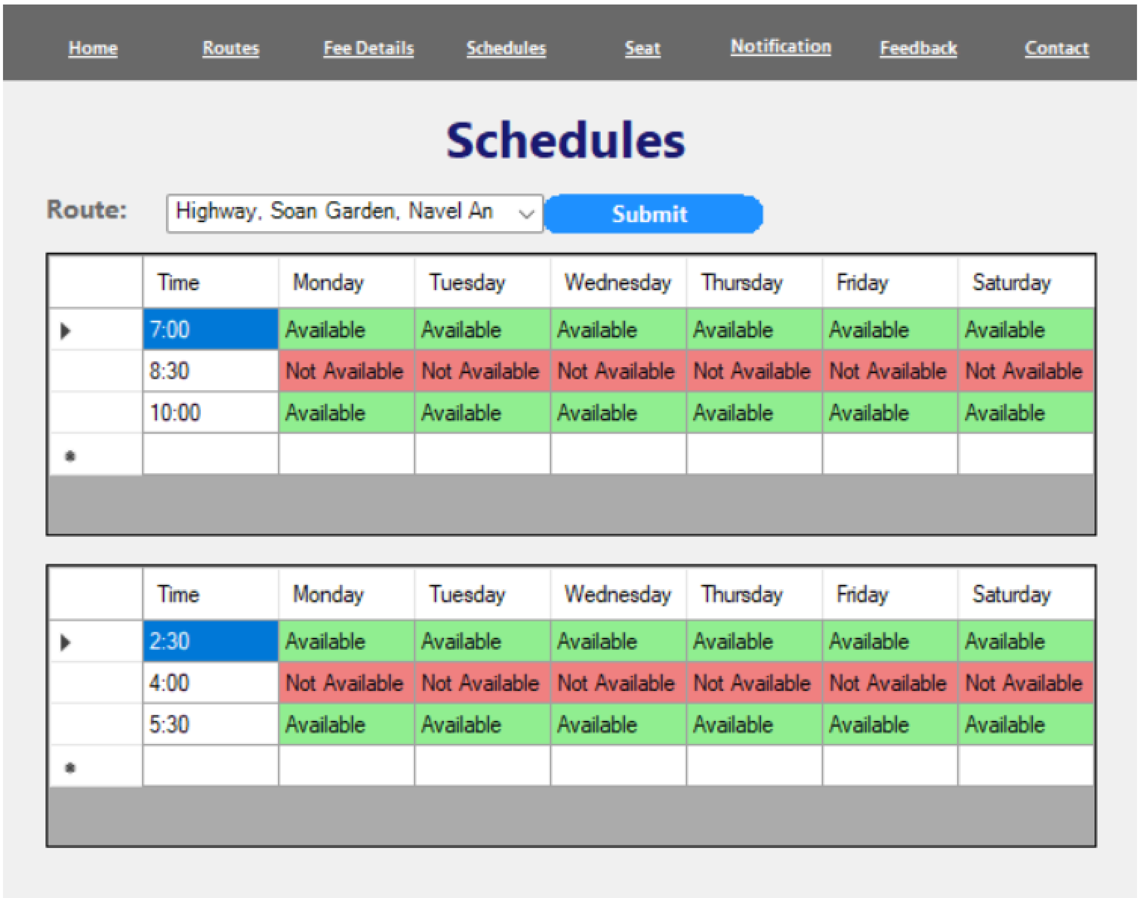

Figure 5.13: Schedules Page

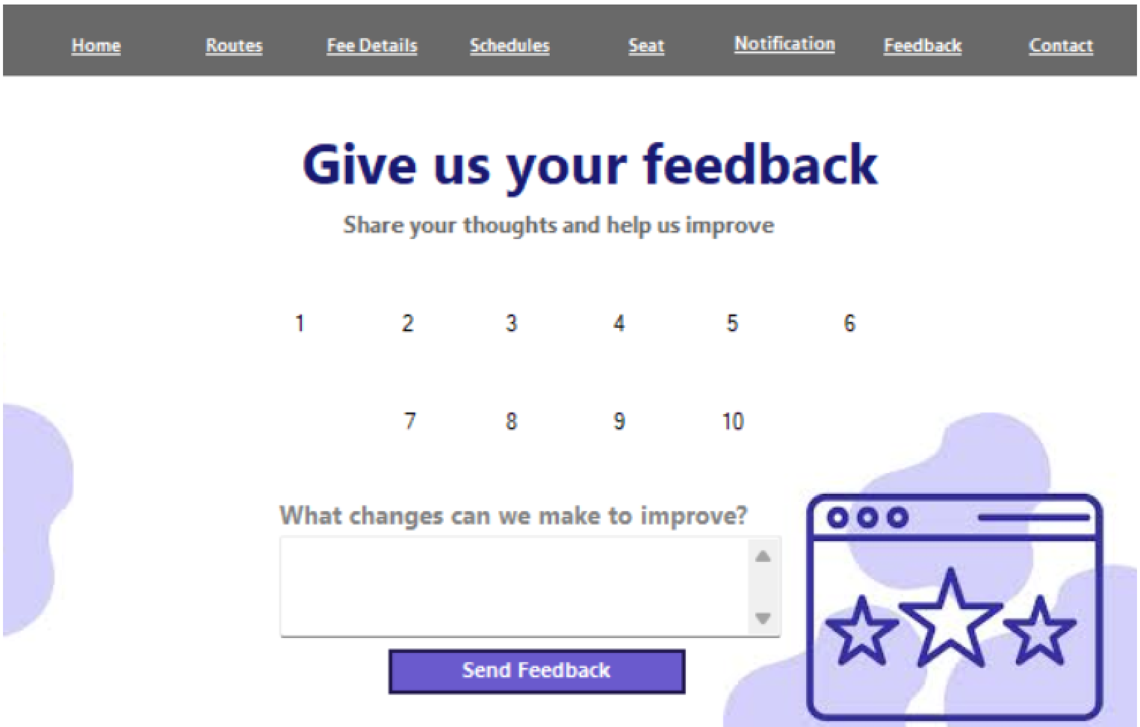

Figure 5.14: Feedback Page

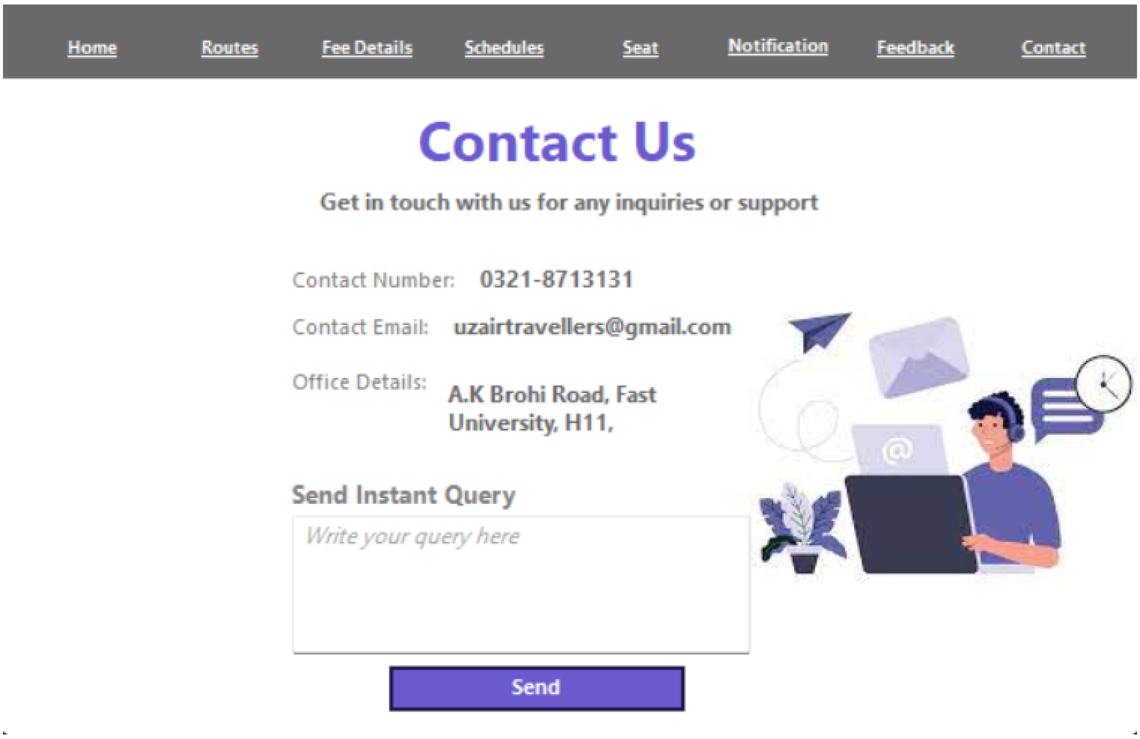

Figure 5.15: Contact Us Page

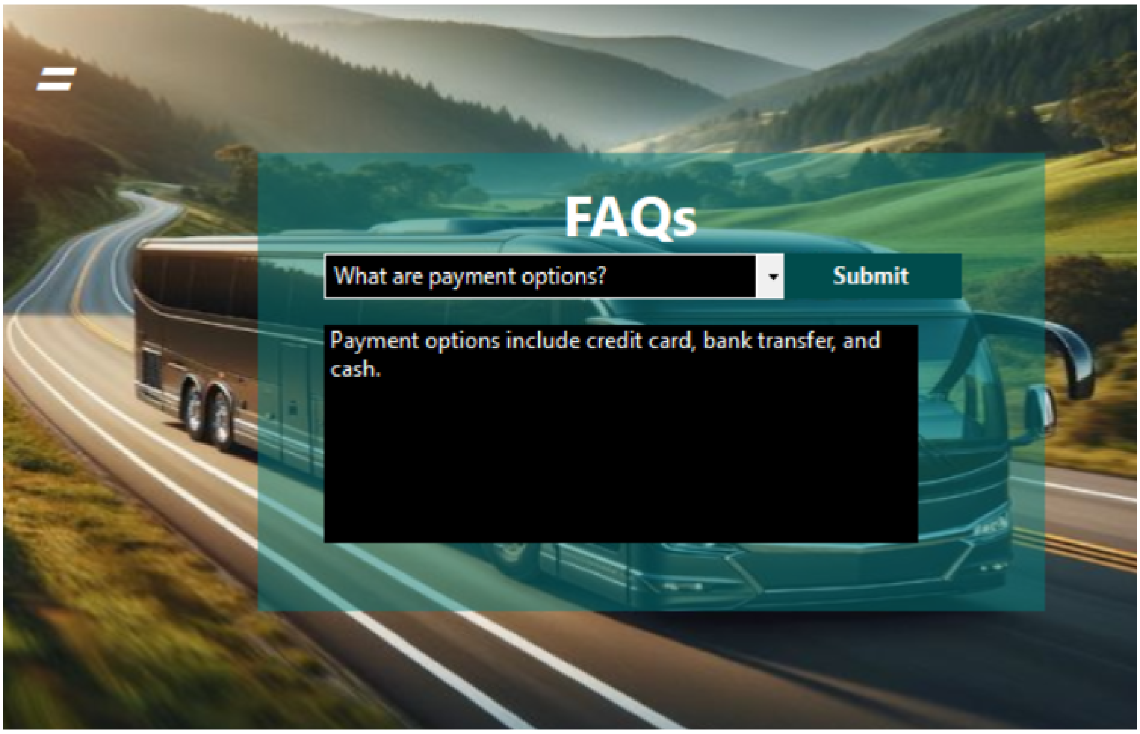

Figure 5.16: FAQ's Page

Chapter 6

Introduction to Product Burn Down Chart

A **Product Burn Down Chart** is a visual tool used in agile project management to track the progress of work completed versus the work remaining in a product backlog over time. It helps teams monitor their progress during a sprint or project and provides a clear view of whether the project is on track to meet its deadlines.

The chart typically shows:

– **X-axis**: Time (usually in days or sprints).

– **Y-axis**: Remaining work (measured in story points, hours, or other units).

As work is completed, the chart "burns down" towards zero, indicating that the team is progressively finishing tasks. A healthy burn down chart shows a consistent decline, signaling that the team is working efficiently. It can also highlight any issues or delays, giving the team a chance to adjust before it's too late.

Project BurnDown Chart

Sprint ID	Sprint Name	Total Features	Completed Features	Remaining Features	Ideal Remaining
0	Initial Planning	16	0	16	16
1	Authentication and Dashboard	16	6	10	12
2	Core Pages	16	6	4	8
3	Remaining Core Pages, UI linking/completion,Backend	16	4	0	4

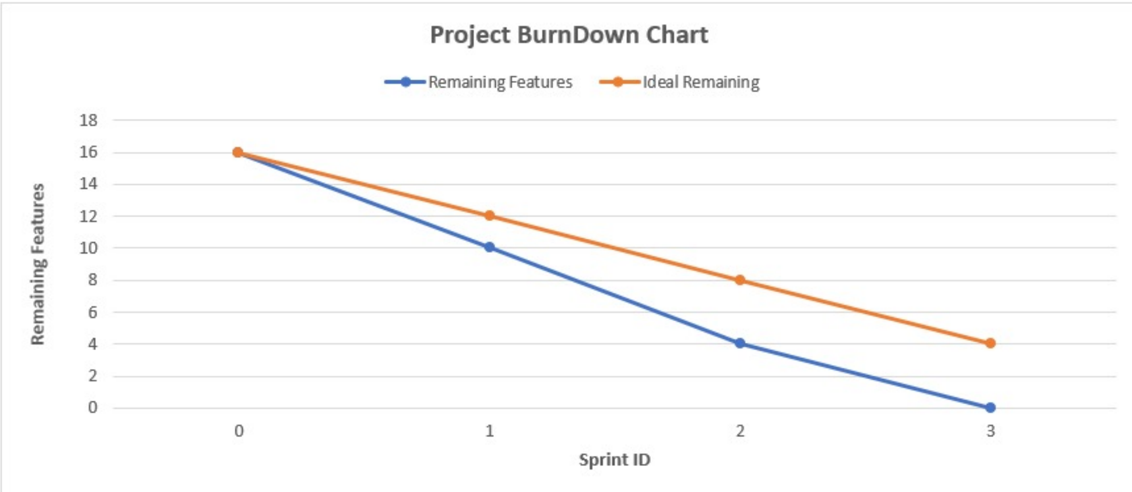

Figure 6.1: Project Burn Down Chart Showing Project Remaining Features over time

Chapter 7

Sprint Workflow Management via Scrum Board

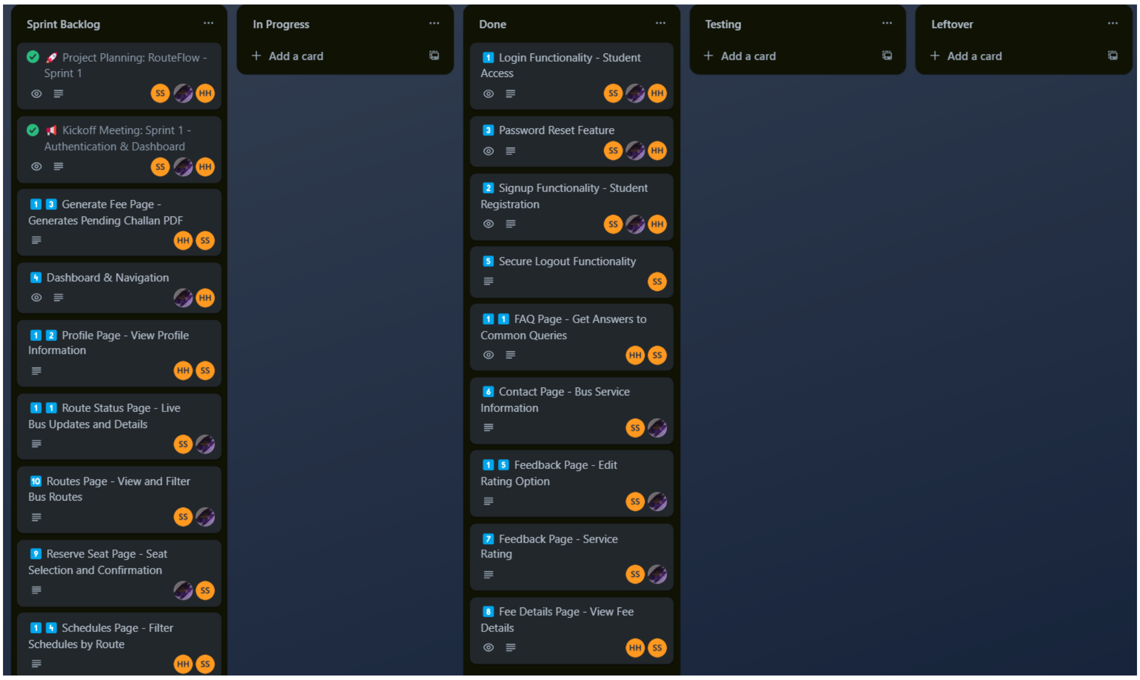

Figure 7.1: Sprint Planning Phase: Task Assignment and Backlog Setup

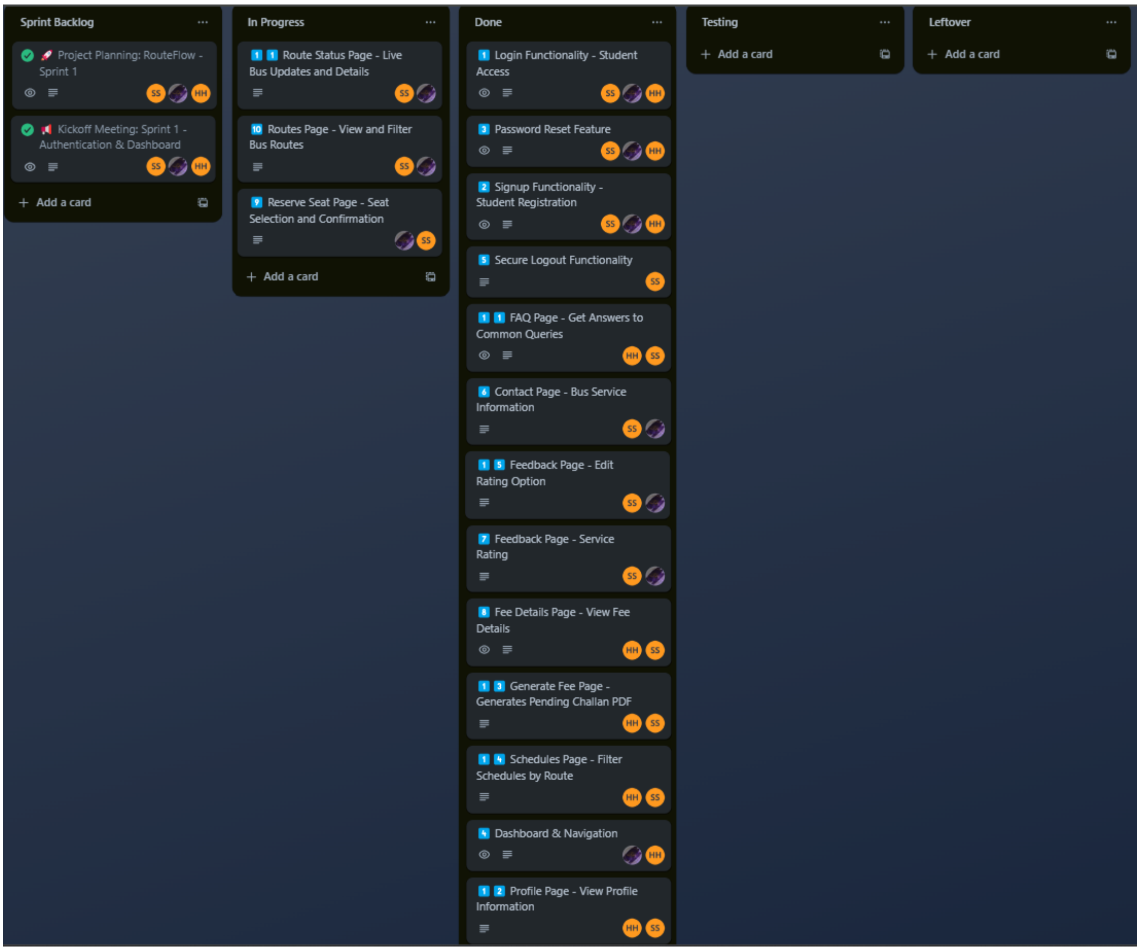

Figure 7.2: Mid-Sprint Progress: Active Development and Task Movement

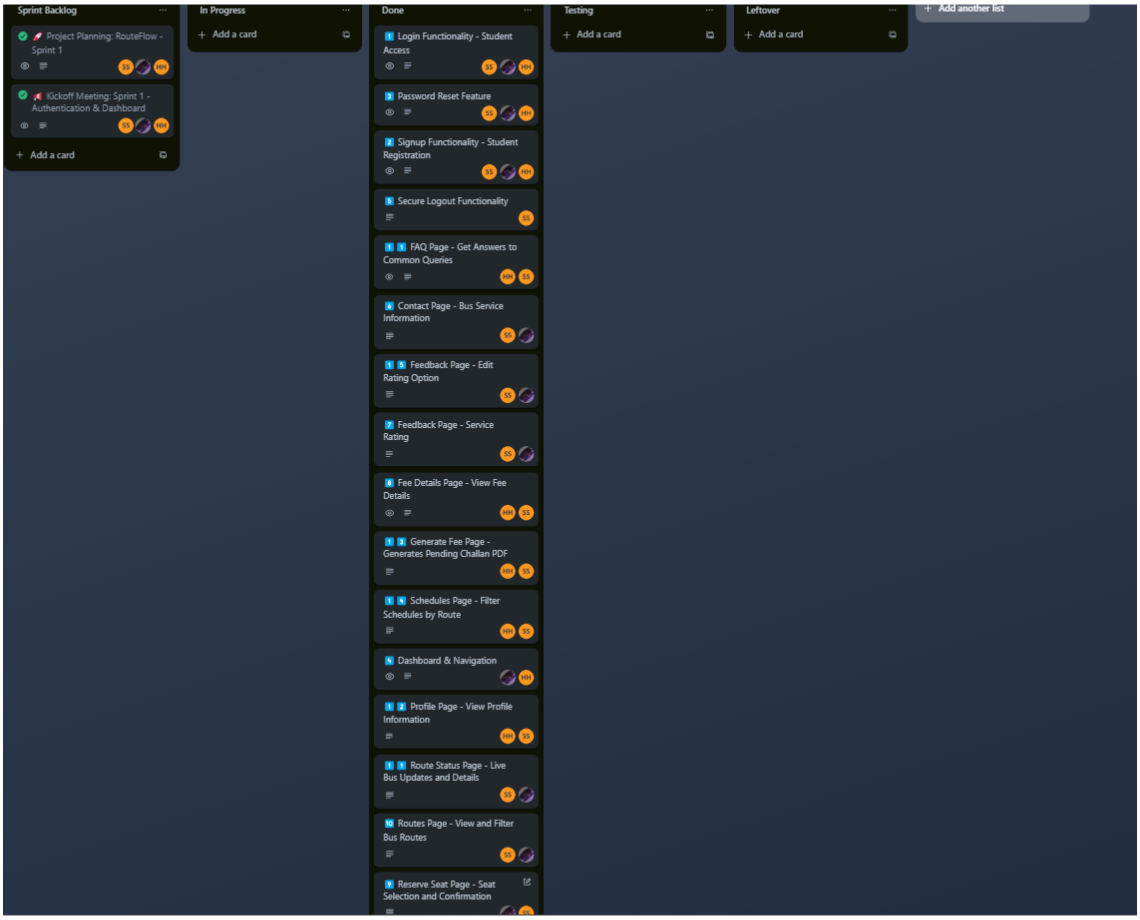

Figure 7.3: Sprint Completion: Task Completion and Testing Phase

Chapter 8

Boundary Value Analysis

8.1 Full Name Field Validation

Table 8.1: Boundary Value Analysis for Full Name Field (2–255 alphabetic characters)

Field	Input Value	Valid/Invalid	Expected Outcome
Full Name	A	Invalid	Reject (too short)
Full Name	Al	Valid	Accept (exactly 2 characters)
Full Name	254-letter alphabetic string	Valid	Accept
Full Name	255-letter alphabetic string	Valid	Accept (upper boundary)
Full Name	256-letter alphabetic string	Invalid	Reject (too long)
Full Name	John123	Invalid	Reject (contains numbers)
Full Name	John Doe (with space)	Valid	Accept

8.2 Roll Number Field Validation

Table 8.2: Boundary Value Analysis for Roll Number (Format: 'i' followed by 6 digits)

Field	Input Value	Valid/Invalid	Expected Outcome
Roll No	i12345	Invalid	Reject (only 6 characters)
Roll No	i123456	Valid	Accept (correct format)
Roll No	I123456	Invalid	Reject (capital 'I' instead of lowercase 'i')
Roll No	i12345a	Invalid	Reject (non-digit character)
Roll No	j123456	Invalid	Reject (doesn't start with 'i')
Roll No	i1234567	Invalid	Reject (8 characters, too long)

8.3 Password Field Validation

Table 8.3: Boundary Value Analysis for Password (4–8 alphanumeric + special characters)

Field	Input Value	Valid/Invalid	Expected Outcome
Password	abc	Invalid	Reject (too short)
Password	abcd	Valid	Accept (exactly 4 characters)
Password	abc123	Valid	Accept (alphanumeric)
Password	abcdef12	Valid	Accept (exactly 8 characters)
Password	abcdef123	Invalid	Reject (too long)
Password	abc@123	Valid	Accept (contains special character)

Chapter 9

Work Division

The tasks in this sprint were equally distributed among the three team members, ensuring an efficient and balanced workflow:

Table 9.1: Team Responsibilities and Contributions

Team Member	Responsibilities and Contributions
Haider Abbas	– HomePage UI fix – Documentation Leadership – Presentation Preperation
Hamd-Ul-Haq	– Schedules Page – Trello Scrum Board management – UI modification and page linking – Project Burndown Chart
Saif Shahzad	– Remaining Backend Logic Work – Boundary Value Analysis. – Collaboration on LaTeX documentation

This structured division of tasks allowed for seamless collaboration, ensuring that each aspect of the project was handled effectively while maintaining agility in development.

Chapter 10

Lessons Learned from the Project

Throughout the course of this project, several valuable lessons were learned, not only about technical concepts but also about managing complexity, adapting to new challenges, and improving the process.

10.1 Importance of Planning and Structure

From the initial stages of creating a complex system (like the university transport management or traffic signal management), I learned that a well-structured plan is essential. This includes breaking down the entire system into manageable components, defining clear user stories, and maintaining an organized workflow. The use of Scrum methodology, like sprint planning and backlogs, helped keep the project on track and ensured smooth execution.

10.2 Adaptability and Problem Solving

As I transitioned between various tasks—whether it was managing data within a database table or Generating Fee challan through pdf. Problems can arise unexpectedly, especially when integrating new features or optimizing existing ones. Flexibility in approaching problems and being able to pivot in the face of challenges was a critical skill.

10.3 User-Centric Design

While working on user interface designs, I realized the importance of creating a user-friendly experience. Whether it was a web-based interface for the event management system or a Windows Forms app for the university transport system, user interaction should always be intuitive. Taking feedback from test users helped refine features and ensure that the final product met user expectations.

10.4 Collaboration and Communication

Effective communication was essential throughout the project. Whether collaborating with a team on system design or presenting the progress in Scrum meetings, ensuring everyone is on the same page, sharing insights, and receiving feedback made the process much more efficient. Having clear communication helped avoid misunderstandings and ensured the team remained aligned.

10.5 Testing and Debugging

The importance of thoroughly testing every module in isolation and as part of the larger system cannot be overstated. Identifying issues early during testing, especially when working with dynamic systems, saved significant time and effort later. Debugging is often a painful process, but a systematic approach—like isolating problem areas and testing each component individually—made this easier.

10.6 Continuous Improvement and Learning

The importance of thoroughly testing every module in isolation and as part of the larger system cannot be overstated. Identifying issues early during testing, especially when working with dynamic systems, saved significant time and effort later. Debugging is often a painful process, but a systematic approach—like isolating problem areas and testing each component individually—made this easier.